

## Parainfluenza Virus type 1, 2 and 3 Nucleic Acid Diagnostic Kit (PCR-Fluorescence Probing)

### Product Identification

Product Name: Parainfluenza Virus type 1, 2 and 3 Nucleic Acid Diagnostic Kit (PCR-Fluorescence Probing)

Reference Number: S3149E-24, S3149E-48,S3149E-24-P

Package Specification: 24 tests/kit, 48 tests/kit, Pre-packaged 24 tests/kit

### Intended Use

The Parainfluenza Virus type 1, 2 and 3 Nucleic Acid Diagnostic Kit (PCR-Fluorescence Probing) is used for qualitative detection of parainfluenza virus types 1, 2 and 3 in the oropharyngeal swabs.The results can be used for auxiliary diagnosis of respiratory tract pathogen infection and provide molecular diagnostic basis for respiratory tract pathogen infection<sup>[1][2][3]</sup>.

Parainfluenza viruses (PIV) belong to the Paramyxoviridae, a subfamily of Paramyxoviridae. According to serotypes, Parainfluenza viruses can be divided into 4 different serotypes, namely parainfluenza virus 1, Parainfluenza virus 2, Parainfluenza virus 3, Parainfluenza virus 4, Parainfluenza virus 1 and Parainfluenza virus 3 belong to the Respirovirus family<sup>[4]</sup>. Parainfluenza 2 and 4 are part of the Rubulavirus genus<sup>[5]</sup>. Parainfluenza virus type 1~3 is an important pathogen causing upper respiratory tract infection, pneumonia and bronchopneumonia, especially in infants<sup>[6][7]</sup>.Now the main methods for laboratory detection of respiratory pathogens include virus isolation and identification, RT-PCR, colloidal gold and etc<sup>[8]</sup>. The test results of this kit are only for clinical reference and shall not be used as the sole standard for clinical diagnosis.

The kit is intended for use by professional, qualified, trained clinical laboratory personnel specifically instructed and trained in the techniques of real-time PCR and *in vitro* diagnostic procedures.

***For in vitro diagnostic use only. For professional use only.***

### Test principle summary and explanation

#### Summary

The Parainfluenza Virus type 1, 2 and 3 RNA is generally detectable in respiratory specimens during the acute phase of infection. Positive results are indicative of the presence of Parainfluenza Virus type 1, 2 and 3 RNA; clinical correlation with patient history and other diagnostic information is necessary to determine patient infection status. Positive results do not rule out bacterial infection or co-infection with other viruses. The agent detected may not be the definite cause of disease. Negative results do not preclude Parainfluenza Virus type 1, 2 and 3 infection. Negative results must be combined with clinical observations, patient history, and epidemiological information.

#### Test Principle

This kit uses specific primers and specific fluorescent probes designed for the conserved region of the nucleic acid of the pathogen to be detected, and is equipped with PCR Mix and other components. By applying real-time fluorescence quantitative PCR technology, on the fluorescence quantitative PCR instrument, this diagnostic kit achieve detection of Parainfluenza Virus type 1, 2 and 3 nucleic acids through the changes of the fluorescent signals.

An internal control targeting the *GAPDH* gene monitors the sample collection, sample handling and qPCR process to avoid false-negative results.

#### Materials provided

This kit is an amplification reaction reagent and contains the following components:

| No. | Reagent Name          | Spec. & Qty.          |                       |                        | Main Ingredients                                                                                   |
|-----|-----------------------|-----------------------|-----------------------|------------------------|----------------------------------------------------------------------------------------------------|
|     |                       | 24 T                  | 48 T                  | Pre-packaged 24T       |                                                                                                    |
| 1   | PIVs-PCR Mix          | 624 μL/tube x 1 tube  | 1248 μL/tube x 1 tube | 26 μL/ tube × 24 tubes | Primers, Probes,dNTPs, MgCl <sub>2</sub> , PCR buffer,                                             |
| 2   | PIVs-Enzyme Mix       | 96 μL/tube x 1 tube   | 192 μL/tube x 1 tube  | 4 μL/ tube × 24 tubes  | Reverse Transcriptase, Taq DNA Polymerase                                                          |
| 3   | PIVs-Positive Control | 1000 μL/tube x 1 tube | 1000 μL/tube x 1 tube | 1000 μL/tube × 1 tube  | Pseudovirus particles containing target genes and internal control gene fragments ( <i>GAPDH</i> ) |
| 4   | PIVs-Negative Control | 1000 μL/tube x 1 tube | 1000 μL/tube x 1 tube | 1000 μL/tube × 1 tube  | Normal Saline                                                                                      |

#### Materials required but not provided

- All contents in this package are prepared and validated for the intended testing purpose. Replacement or modification of any of the package contents will affect the testing performance of the kit. Components contained within a kit are intended to be used together.
- All biological samples in this diagnostic kit have been inactivated.
- Materials required but not provided: 1.5mL DNase-free and RNase-free microfuge tubes; 0.2mL PCR reaction tubes or strip; Various models of pipettes and pipette tips (10μL, 200μL and 1000μL tips with filters); Microcentrifuge; Vortex mixer.
- Reagent required but not provided: Sample Release Reagent ( Reference Number : S1014E Series) manufactured by Sansure Biotech Inc. and Nucleic Acid Extraction-Purification Kit (Reference Number: S50016E Series) manufactured by Sansure Biotech Inc.

#### Warnings and precautions

##### Warnings

- Do not mix or exchange components from different kits.
- All biological materials in the kit have been inactivated.

##### Precautions

- For *in vitro* diagnostic use only. Please read the product manual carefully before operation.
- Please learn and be familiar with the operation procedures and precautions for each instrument before test. Please make sure quality control is performed for each test.
- Laboratory management shall strictly follow management practices of PCR gene amplification laboratory, laboratories personnel must receive professional training, test processes must be performed in separated regions, all consumables should be for single use only after sterilization, special instruments and devices should be used for every process, all lab devices required in different processes and regions should not be cross-used.
- All samples for detection should be handled as potentially infectious. Wear laboratory coats, protective disposable gloves and change the gloves often to avoid cross-contamination between samples. Handling of samples and waste must meet relevant requirements outlined in local, state and national regulations.
- Note: Improper operation during the storage, transportation and use of the reagent may affect the test results. For example, improper storage and transportation, sample collection, sample processing and test process are not standardized. Please strictly follow the instructions.
- Due to the characteristics of swab and other sample collection process and viral infection process itself, false negative results may be caused by insufficient sample volume, which should be combined with other clinical diagnosis and treatment information for comprehensive judgment, retest when necessary.
- After the addition of the sample in the tube the resulting solution is to be considered potentially biohazardous, handle the reagent with appropriate precautions and good laboratory practice.
- The safe disposal of the reagents supplied must be carried out according to the instruction contained in the specific Safety Data Sheets and in compliance with the national regulations on disposal of potentially hazardous waste.
- Any serious incident that has occurred in relation to the device shall be reported to the manufacturer and the competent authority of the Member State in which the user and/or the patient is established; If you have any questions about the test or the results, please contact Sansure’s customer service hotline +86-731-88883176-6116 or send an email to info@sansure.com.cn/ support@sansure.com.cn.

#### IVD storage, operating conditions and stability

- This kit should be stored in its original box at -25℃ to -15℃ and protected from light. The kit is valid for 12 months.
- Please refer to the date of manufacture and expiry date printed on the outside of the box.
- Unopened reagents are valid and stable until the expiry date.

- Once the reagents are opened, the maximum number of freeze/thaw cycles should not exceed three.
- The reagents keep valid and stable before the expiry date on the outer package when transporting for 7 days in a sealed foam box containing coolant with the temperature lower than 20℃.

#### Instrumentation

The kit is compatible to Fluorescence Quantitative Analysis System containing FAM, HEX,ROX and CY5 channels such as:

- Applied Biosystems/7500 Real-Time PCR System
- Molarray/Real-Time Quantitative Thermal Cycler (Model: MA-6000 )
- Hongshi/SLAN®-96P Real-Time PCR System
- ThermoFisher/QuantStudio™ 5 Real-Time PCR System
- Sansure /Portable Molecule Workstation (Model: S-Q31A/S-Q31B)
- Sansure/Portable Molecular Workstation (Model: S-Q36A)

#### Specimen Requirements

- Applicable specimen type: Oropharyngeal swab.
- Collection of specimen

**Oropharyngeal swab:** Collect samples in accordance with the relevant local procedures for COVID 19 Laboratory Testing. Collection swabs should have a synthetic tip, such as nylon or Dacron®, and an aluminium or plastic shaft. Calcium alginate swabs are unacceptable and cotton swabs with wooden shafts are not recommended. The specimen collection tube should be pasted with the barcode first; the oropharyngeal swab should be collected within 3 days after the onset of the disease. A sterile flocking swab should be used for sampling, moderately wipe the posterior pharyngeal wall, avoid touching the tongue. Quickly place a sterile swab into the collection tube used for collection of oropharyngeal swabs. Break the sterile swab rod near the top, tighten tube cap and seal with sealing film.

- Storage and delivery of specimens:

Specimens to be tested can be immediately processed, specimens to be tested within 24 hours can be stored at 2℃ to 8℃. Specimens that cannot be detected within 24 hours should be stored at -70℃ or below (in the absence of -70℃ storage conditions, specimens to be tested can be stored at -25℃ to -15℃ for 10 days, nucleic acid can be stored at -25℃ to -15℃ for 15 days). Multiple freeze/thaw cycles should be avoided. Specimens should be transported in a sealed frozen container with ice or in a sealed foam box with ice packs.

#### Test procedure

**1. Please process according to the following steps for SLAN®-96P Real-Time PCR System, 7500 Real-Time PCR System, QuantStudio™ 5 Real-Time PCR System, Real-Time Quantitative Thermal Cycler (Model: MA-6000 ):**

**1.1 Preparation of reagent (performed at “reagent preparation room”)**

- 1.1.1 Take out each component from the diagnostic kit and place them at room temperature. Allow the reagents to equilibrate at room temperature, then vortex each of them respectively for later use.
- 1.1.2 Prepare the PIVs-PCR Master Mix according to following table. The volume required is based on the total number of specimens, plus a PIVs-Positive Control and a PIVs-Negative Control. Mix thoroughly then centrifuge it for later use. The remaining reagent must be stored at -20℃ immediately.

|                      | 1 sample | 10 samples | 24 samples | 48 samples |
|----------------------|----------|------------|------------|------------|
| PIVs-PCR Mix (μL)    | 26       | 260        | 624        | 1248       |
| PIVs-Enzyme Mix (μL) | 4        | 40         | 96         | 192        |

Note: The above configuration is for reference only.

**1.2. Processing and loading of specimens (performed at “specimen processing room”)**

1.2.1 Use Nucleic Acid Extraction-Purification Kit( Reference Number: S50016E Series ) manufactured by Sansure Biotech Inc. to extract the nucleic acid according to corresponding manual.

1.2.2 Add 20 μL of the extracted RNA to the PCR tubes in the following order: PIVs-Negative Control, patient specimen(s), and PIVs-Positive Control.

1.2.3 Add 30 μL of PIVs-PCR Master Mix into each well. Cover each well and centrifuge at 2000 rpm for 10 seconds, place into the Real-Time PCR system.

**1.3 PCR Amplification** (Refer to user manual of each instrument to adjust the settings.)

1.3.1 Place PCR reaction tubes into the specimen wells of the amplification equipment. Set up the PIVs-Positive Control, PIVs-Negative Control and specimens to be tested in order and input specimen name.

1.3.2 Select PCR test channel:

- Select FAM channel to test Parainfluenza Virus type 3 nucleic acid.
- Select HEX/VIC channel to test Parainfluenza Virus type 2 nucleic acid.
- Select ROX channel to test Parainfluenza Virus type 1 nucleic acid.
- Select CY5 channel to test internal control.

1.3.3 Set cycle parameters

|   | Steps                                           | Temperature | Time     | Cycle No. |
|---|-------------------------------------------------|-------------|----------|-----------|
| 1 | Reverse transcription                           | 50℃         | 10 min.  | 1         |
| 2 | cDNA pre-denaturation                           | 95℃         | 1 min.   | 1         |
| 3 | Denaturation                                    | 95℃         | 15 sec.  | 45        |
|   | Annealing, extension and fluorescence detection | 60℃         | 30 sec*. |           |

**\*Note:** This cannot be set to 20 seconds due to ABI 7500 instruments, but can be set to 31 or 32 seconds.

When the settings are completed, save the settings and carry out the reaction procedure.

**2. Please process according to the following steps for Portable Molecule Workstation ( Model: S-Q31A&B ):**

#### 2.1 Preparation of consumables and reagents

2.1.1 Take out the reaction tube carrier, PCR reaction tube and Tip.

2.1.2 Put the **Tip** into **Well H**,and **PCR reaction tube** into**Well I** PCR(The well location information has been marked on the reaction tube carrier).

2.1.3 Put Sample Release Reagent (Reference Number : S1014E Series) into the **Well B**; Put PIVs-PCR Mix into the **Well I** C; Put PIVs-Enzyme Mix into the **Well D**.

2.1.4 Add 20μL sample to be tested or PIVs-Positive Control or PIVs-Negative Control into the **Well B** (To avoid bubbles during operation, it is recommended to pipet deeply and release slowly ).

#### 2.2 Test Procedure

2.2.1 Gently press the front door to open it.

2.2.2 Place the **Well A** of reagent strip into the instrument towards the outside of the instrument,and close the front door of the instrument.

2.2.3 Click the"**Lab task**" on the instrument display screen to enter the interface of setting new experimental task.

2.2.4 Select the required experimental project in the drop-down menu of **Lab project**, enter the corresponding task name in the **Task Name** bar, and input and select other items that should be input or selected.

2.2.5 Click "**Submit**" to submit the experimental task and "**OK**" to run the instrument and start the experimental task successively.

2.2.6 When the Portable Molecule Workstation ( Model: S-Q31B ) shows "**Please transfer the PCR tube to the 1/2/3/4**"(The S-Q31A shows "Please transfer the PCR tube") on the interface, take out the PCR tube and cover it well, then centrifuge it instantaneously.

2.2.7 Insert the PCR tube into the PCR amplification module ( the "PCR 1/2/3/4" cover has been automatically opened at this time ), close the PCR lid of the amplification module, then click "**OK**" for amplification detection.

3. Please process according to the following steps for Portable Molecular Workstation ( Model: S-Q36A ):

3.1 Preparation of consumables and reagents

3.1.1 Take out the Consumables kits and reagents.

3.1.2 Put Sample Release Reagent (Reference Number : S1014E Series) into the **Well B**; Put PIVs-PCR Mix into the **Well C**; Put PIVs-Enzyme Mix into the **Well D**.

3.1.3 Add 20μL sample to be tested or PIVs-Positive Control or PIVs-Negative Control into the **Well A** (To avoid bubbles during operation, it is recommended to pipet deeply and release slowly ).

3.2 Test Procedure

3.2.1 Click the "and"" button on the instrument display screen to open the door of the instrument and put the prepared consumables into the designated position of the instrument.

3.2.2 Select the required experimental project in the drop-down menu of **Lab project**, enter the corresponding task name in the **Task Name** bar, and input and select other items that should be input or selected.

3.2.3 Click **"Submit"** to submit the experimental task and **"OK"** to run the instrument and start the experimental task successively.

Reading test results

1. **Result Analysis** (Refer to user manual of instrument to adjust the settings.)

Results will be saved automatically when reactions are completed. Analyze amplification curve of target of detection and internal control. Adjust Start, End and Threshold values of Baseline of the graph according to analysis result (Users can adjust the values according to the actual situation. Start value can be set between 3-15, and End value between 5-20. Adjust the amplification curve of negative control to be flat or below threshold). Click "Analysis" to implement the analysis, make sure each parameter satisfies the requirements given in **"5. Quality Control"**. Go to "Plate" window to record qualitative results.

2.Quality Control

All test controls should be examined prior to interpretation of patient results. If the controls are not valid, the patient results cannot be interpreted. The following describes the result interpretation concerning the use of the controls mentioned above.The Ct cutoff value of this kit is set as 40 and the end user is required to review fluorescent curves before final interpretation. For weakly positive samples, all positive curves are typical S-shaped amplification curves or platform-free periods.

5.1 PIVs-Negative Control:The FAM ,HEX/VIC ,ROX and internal control(CY5) channels are negative, no Ct or Ct>40.

5.2 PIVs-Positive Control: The FAM ,HEX/VIC ,ROX and internal control(CY5) channels have the typical s-type amplification curve,the Ct≤35.

5.3 The test result is treated as valid if all the conditions in the above-mentioned are met for the same test. Otherwise the test result is treated as invalid and needs to be re-tested.

Note: Result of (-): Ct value >40 or Undetermined; Result of (+): Ct value ≤ 40.

If there is contamination for the re-test, please perform decontamination procedures.

Reference Range

Through the research on reference values, the Ct reference value of target gene is determined to be 40, the Ct reference value of internal control is determined to be 40.

Explanation of Detection Result

Assessment of clinical specimen test results should be performed after the positive and negative controls have been examined and determined to be valid and acceptable. If the controls are not valid, the patient results cannot be interpreted. The end user is required to review fluorescent curves before final interpretation.For weakly positive samples, all positive curves are typical S-shaped amplification curves or platform-free periods (38≤Ct≤40).

1. For samples with a typical S-type amplification curve detected in the FAM channel and Ct≤40 ,it means that the result of Parainfluenza Virus type 3 test was positive; For samples with no typical S-type amplification curve detected in FAM channel or with Ct > 40 , the CY5 channel has an amplification curve and Ct≤40 , ,it means that the result of Parainfluenza Virus type 3 test was negative.

2. For samples with a typical S-type amplification curve detected in the HEX/VIC channel and Ct≤40 ,it means that the result of Parainfluenza Virus type 2 test was positive; For samples with no typical S-type amplification curve detected in HEX/VIC channel or with Ct > 40 ,the CY5 channel has an amplification curve and Ct≤40 , it means that the result of Parainfluenza Virus type 2 test was negative.

3. For samples with a typical S-type amplification curve detected in the ROX channel and Ct≤40 ,it means that the result of Parainfluenza Virus type 1 test was positive; For samples with no typical S-type amplification curve detected in ROX channel or with Ct > 40 , the CY5 channel has an amplification curve and Ct≤40 , ,it means that the result of Parainfluenza Virus type 1 test was negative.

4. For FAM ,HEX/VIC, ROX, and CY5 channels, no typical S-type amplification curve (No Ct) is detected, or Ct>40, which means that the cell content of the test sample is too low or there is an interference substance that inhibits the reaction. The test of this sample If the result is invalid, the reason should be found and eliminated, and the sample should be re-sampled for repeated tests (if the test results of the repeated tests are still invalid, please contact our company).

Result of (-): Ct value >40 or Undetermined; Result of (+): Ct value ≤ 40. If the retest is contaminated, please carry out purification treatment.

For technical assistance of test results, please contact local distributor or email: info@sansure.com.cn.

Limitations of the procedure

1. False negative results can be caused by poor specimen quality, improper sample collection, improper transportation, improper laboratory processing, or a limitation of the testing technology.
2. Mutation in the target sequence of Parainfluenza Virus type 1, 2 and 3 or change in the sequence due to virus evolution may lead to false negative results.
3. Improper reagent storage may lead to false negative results.
4. Use of this assay is limited to personnel who are trained in the procedure.
5. Test results of the diagnostic kit can only be used as an aid in clinical diagnosis. Symptoms and physical signs, disease history, other laboratory examinations and therapeutic reactions of the patients should be comprehensively considered for the clinical diagnosis and treatment.
6. Unverified interfering substances or PCR inhibitors may lead to false negative or invalid results.

Performance characteristics

1. Inclusivity

Based on the specific sequences of Parainfluenza Virus type 1, 2 and 3 retrieved from NCBI database, comparative analysis of computational sequences was conducted for the inclusion of primers/probes used in the Kit. The comparative analysis showed that the target sequences of Parainfluenza Virus type 1, 2 and 3 from different sources and different strains had 100% homology.

2. Specificity

There was also no cross-reaction with common respiratory pathogens(The positive specimen of coronavirus, Bocavirus,Metapneumovirus,Influenza A virus, Influenza B virus Yamagata type, Influenza B virus Victoria type,Influenza A virus H1N1 type,Influenza A virus H3N2 type, Influenza A virus H5N1 type, Influenza A virus H7N9 type,Respiratory syncytial virus A and B,Rhinovirus A , B and C,Adenovirus 1 , 2 , 3 , 4 , 5 , 7 and 55 type,Enterovirus (71 type) ,Coxsachievirus A,Measles virus,Mumps virus,Acinetobacter baumannii,Streptococcus pneumoniae,Haemophilus influenzae,Pseudomonas aeruginosa,Legionella pneumophila,Bordetella pertussis,Staphylococcus aureus,Mycoplasma pneumoniae,Chlamydia pneumoniae,Candida Albicans,Mycobacterium tuberculosis,Nontuberculosis mycobacteria).(The above pathogens are all verified at the concentration level of 10<sup>6</sup>copies/ml ~10<sup>8</sup> copies/ml.) The negative Enterprise reference panel are all negative.

3. Limit of detection

The limit of detection of this kit is 400 copies/mL.

4. Precision

The coefficient of variation (CV%) of Ct value of the within-run precision is ≤ 5%.

5. Possible interfering substances in specimens

100ug/mL hydroxymezoline hydrochloride, 50ug/mL dexamethasone, 50ug/mL cefmetame hydrochloride, 100ug/mL oseltamivir, 100ug/mL zanamivir, 100ug/mL ribavirin, 100ug/mL azithromycin, 50ug/mL fluticasone, 100ug/mL peramivir, 50ug/mL budesonide, 60ug/mL sodium chloride, 100ug/mL urea, 10ug/mL heme, 20ug/mL purified mucin have no significant interference with the detection results of the kit.

Doc.:S3149E IVDD IFU V02

List of references

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7. Evaluation of a Live-Attenuated human Parainfluenza Type 1 Vaccine in Adults and Children. J Pediatric Infect Dis Soc. 2015 .
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Symbol key

| Symbols                                                                               | Meanings                                            | Symbols                                                                               | Meanings                                                                                                                  |
|---------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|    | <i>In Vitro</i> Diagnostic Medical Device           |    | Batch Code                                                                                                                |
|    | Use-by date                                         |    | Reference Number                                                                                                          |
|    | Manufacturer                                        |    | Date of Manufacture                                                                                                       |
|    | Contains sufficient for <n> tests                   |    | Temperature Limit                                                                                                         |
|    | Caution                                             |    | Consult Instructions for Use                                                                                              |
|    | PAP21: Not corrugated cardboard                     |    | Do not re-use                                                                                                             |
|    | Authorized representative in the European Community |    | This product fulfills the requirements of the European Directive 98/79/EC for <i>in vitro</i> diagnostic medical devices. |
|   | PCR Mix                                             |   | Enzyme Mix                                                                                                                |
|  | Negative Control                                    |  | Positive Control                                                                                                          |
|  | Prepackaging                                        |  | Keep away from light                                                                                                      |
|  | Version                                             |                                                                                       |                                                                                                                           |

Contact information



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